

Rice University
ELEC 494/496- Senior Design
8:30-9:50 am MWF

This course is significantly linked with its counterparts in the other Collaborative Capstone courses: BioEngineering (BioE 451-2), Global Health (GHLT 451-2), MechE (Mech 407-8), and Materials Science (MSNE 407-8). The spring semester of the ECE course is Elec 496. All of the collaborative Capstone courses carry 4 credits in the fall and 3 in the spring

INSTRUCTORS

For specific questions about the course or grading as it pertains to any individual student- the instructor for the course the student is registered for should be consulted first and has the final decision-making authority.

BIOE 451/2 - Bioengineering Design I and II

Dr. Sabia Abidi (sabia.z.abidi@rice.edu)

GLHT 451/2 - Bioengineering Design I and II

Dr. Kiara Lee

ECE 494- Senior Design

Dr. Gary Woods

gary.woods@rice.edu

Office Address: DCH 2023

713-348-3598- (note- no voicemail- email preferred)

Office Hrs: By appointment. I also have regular evening office hours that you can schedule using your Rice Google credentials (*not* personal gmail.)

MECH 407/408 - CAPSTONE DESIGN PROJECT I and II

Dr. David Trevas

trevas@rice.edu

Office Hrs: By appointment

OEDK SUPPORT STAFF

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COURSE INFORMATION

Meeting Info: 8:30-9:50 am MWF
Classroom as noted on schedule

Key Resource information: OEDK website: <http://oedk.rice.edu>
Canvas website: <https://canvas.rice.edu/courses/43826>

Course Textbook: (OPTIONAL FOR STUDENTS: Source of some lecture materials, useful for teams)

Title	Product Design and Development
Author(s)	Karl T. Ulrich and Steven D. Eppinger
Edition, Year	Fifth, 2011
Publisher	McGraw-Hill

Course Summary and Objectives

This class is not like the other courses you have probably taken. There is not a single book or a single concept area or set of equations that you need to master to succeed in this class. Many different topics, tools and ideas will be presented and it is up to you to combine those with your knowledge of engineering to go through the process of designing a useful product. You are expected to work hard, find your own resources, engage in effective teamwork, be creative, be analytical, and be an independent engineer to succeed in this class. You MUST take both semesters of this course in sequence during the same academic year.

A design and analysis project will be scheduled for the duration of both semesters. The process of developing and building the design project is the primary focus of the course. The project's topic will be described in detail in mid-September. Students will be divided into teams. During the year, written progress reports, an oral proposal and progress report, an oral elevator pitch, final poster presentation and final oral reports will be required.

Learning outcomes – By the end of this yearlong course, students will be able to:

- Identify key customer needs for a project
- Identify market opportunities in commercially oriented projects, or the state of current knowledge in research-oriented projects
- Develop specifications for final deliverables
- Brainstorm possible solutions and select final concept
- Utilize engineering knowledge and systems engineering concepts to design an appropriate solution for the project
- Function effectively in a team, providing feedback to peers

- Devise and execute a plan to implement the project
- Communicate at a professional level through oral, visual, and written methods

A student's responsibilities in this course may be divided into three areas: 1) Lectures and workshops on important topics in design, 2) the design process itself, and 3) documentation of the design. Lectures and workshops will occupy about three hours a week. Each student should expect to devote an average of at least 10 hours per week for the Fall Semester in addition to attending lectures and/or workshops.

Detailed descriptions of the required deliverables will be provided at a later date. Briefly, they will include:

- A weekly meeting of the team with its faculty advisor for part of the year
- A team "Project Binder" containing softcopies of all documentation (reports, presentations, and supporting materials)
- An end-of-semester Oral Presentation (each semester)
- Presentation of the team's prototype at the annual Engineering Design Showcase
- Entering at least one external design competition or submitting at least one conference paper
- A functional prototype that has been tested against design criteria

At the end of the course students will have experience taking a project from initial problem definition to working prototype, following the steps of the engineering design process.

BIOE, MECH, & ELEC Students in the Global Health Technologies Minor: In order to meet the requirements of the GLHT Minor, you must participate on a Global Health related design project.

COURSE ASSIGNMENTS

Each of the 3 courses has somewhat different assignments and grading policies. Each project will have a "home" department and all team members will be responsible for the assignments from the associated class, regardless of which department the students are in. For example, if an Elec student is on a BioE project, then all team members including the Elec students should submit assignments according to the BioE course (BioE 451-2). Consult the corresponding syllabus or ask the appropriate course faculty if you have questions about the requirements.

Homework/ Pre-Class Preparation

For the majority of lectures, no pre-class work is required. It is always helpful to read the material in the book as much more is covered in the text than will be covered in lecture. Occasionally, students will be asked to prepare material prior to class for use in their team sessions. In such cases, a printed copy of the preparation *will be due at the beginning of class lecture on the day it is due*. No late homework will be accepted without permission of the instructor **before** the homework is due. Illness and family emergencies will be dealt with on an individual basis.

Exams

There will be no final exam, but there will be a final presentation during final exam periods in each semester and a written report will be due during finals week.

Projects

Students will complete a design and analysis project that will run for both semesters. The process of developing and building the design project is the primary focus of the course. The project's topic options will be described in the first week of class. Students will be divided into teams. Some of the homework assignments will be designed to help students with their design project. During the year, teams will complete various oral and written communication assignments as described in the course schedule posted online.

Intellectual property

Some projects have external sponsors who retain rights to intellectual property generated by students working on these projects. Students assigned to these projects will need to sign a contract assigning their IP rights to the sponsor and also providing for non-disclosure of confidential information provided by the sponsor. Such "IP" projects will be clearly noted in the project menu. No student is required to apply for an IP project. There will be several project options where Rice or the students would be expected to own any IP rights arising from the project, in accordance with Rice policies.

Interdisciplinary projects

Some teams will be composed of students from multiple departments. Those teams will be designated as belonging to one of the 4 classes and should follow the assignments for that class. For example, a team of 2 Elecs + 2 BioEs + 1 Mech working on a medical-device project might be designated as a BioE team, and will be expected to fulfill the requirements for BioE 451-2. There could be a few department-specific assignments that the Mechs & Elecs would need to complete, independent of the BioE team requirements. Dr. Woods will make the final assignment of grades for the Elec students based on the recommendation of the faculty of the course to which the team has been designated.

Data Science projects

Students will have the option of participating in projects that are administered by the Data Science capstone course, DSCI 435. Students should coordinate with the instructor of that course, Dr. Arko Barman, as well as with Dr. Woods. Generally speaking a DSCI team will complete the DSCI assignments plus a few ELEC 494 assignments. The course grade is still assigned by Dr. Woods on the recommendation of the DSCI professor.

Peer review surveys

Students are required to fill out the CATME online peer evaluation survey twice in each semester. Failure to fill out the CATME evaluation (including the free-response sections) will result in an individual-grade reduction as described below.

Students are also encouraged to participate in one or more online research surveys being conducted by the Psychology Dept. Students will need to complete the surveys for course purposes but will have the option of not having their responses used in the research. In any event the data collected will remain strictly confidential. Students who wish to be excluded from the surveys entirely should contact Dr. Woods to arrange an exception.

COURSE ADMINISTRATION

Grading Policy - The final grade will be based on the following.

4

Course Syllabus

BIOE 451/2, ELEC 494/6, MECH 407/8, GLHT 451-2, MSNE 407-8

Fall 2025-Spring 2026

- Team project grade -- 90 %.
 - Each team member by default receives the team's score as a baseline.
 - items contributing to team score (see Design Documents Grading sheet online, subject to change and varying somewhat depending on whether the team is mentored by BioE, GLHT, Mech, or Elec faculty)

	Semester 1	Semester 2
● Oral Presentations	~17%	~9%
● Design Execution/Prototypes	~41%	~56%
● Written Documentation	~42%	~36%

- The team portion of each individual's grade is adjusted up & down based on individual performance. Factors affecting individual grades:
 - Attendance in class (see below)
 - Performance in oral presentations as assessed by course professor
 - Attendance and participation in team meetings as assessed by course professor
 - Contribution to team as assessed in CATME peer evaluations and by course professor
- Individual assignments – 10%.
Occasionally there may be individual assignments for ABET-required items. For these assignments, each student should submit his/ her individual work. Failure to submit the work could result in an incomplete grade.

Attendance

The following lecture attendance policy applies for most of the lectures.

- Students may miss 2 classes per semester for any reason and will not be penalized
- Students who have job interviews, medical school or graduate school interviews will have those absences count as ½ of an absence. However, these must be cleared with the instructor at least 2 days AHEAD of time.
- Arriving late to class will count as ½ absence.
- Extended illness that will keep you out of class for more than 2 days will be dealt with on a case by case basis with the instructor
- After the first 2 absences, your grade is reduced 1.5 % per absence for the next 2 absences and then 3% per absence after that.
- It will be considered an honor code violation to sign in another student for attendance or to have a student sign you in to this class. The trouble that would come from this behavior is not worth it for EITHER student.
- If you need allowances due to your location or lack of internet or similar reasons, please discuss this with your instructor.

Weekly meetings

Each team should meet with their course faculty mentor approximately once per week. They will discuss their progress and plans and any obstacles encountered, as summarized in their weekly report. To the extent possible, these meetings will take place face-face in a physically distanced manner somewhere on campus. Remote students should plan to attend by teleconference.

Policies

Class attendance is **essential** in this course. There are no exams. You are expected to complete any assigned pre-class assignments before class, understand the material presented and apply it to your design project. This

is not possible without consistent class attendance. There are many weeks during the 2nd semester where there is no class scheduled. This is to allow more time to complete your design projects.

Any disputes over grading on the homework or reports should be brought to the instructor. Do not take questions related to grading to the TA's. If the dispute involves other than an arithmetic error in your score, the entire report or homework will be re-graded. Re-grading may result in a net loss of points. Submit the original, unaltered, homework or report along with a written description of the potential error. If a classmate's report or homework is being submitted as documentation of the grading error, it is also subject to possible re-grading. Late homework will generally not be accepted except in the case of illness or serious emergency. Contact the instructor before the due date to arrange an acceptable due date. Illnesses and emergencies should be documented with an appropriate excuse.

General Expectations of Conduct with Respect to Course Materials, Supplies and Peer and Staff Assistance (AKA- OEDK Citizenship 101)

OEDK general supplies are available for use by all teams. It is expected that teams will use what they need and not hoard supplies that others can use. If a part or supply is purchased specifically for a team, other teams must ask and get permission to use them for their own project. However, we expect a spirit of collaboration and help for each other. This means that we expect teams to be willing to help each other out with parts and supplies if possible. Remember, all supplies purchased for your project are actually owned by Rice.

Equipment that has been checked out to a particular team is their responsibility and property for the duration of their check-out period. You are expected to check out your own materials and not remove them from another team's table. If it is an emergency... and you use another team's tool- RETURN IT! Better yet check out your own! Open tools should always be returned to their proper location but may be used by any team if they are found on another team's table.

Honor Code Policy

The Rice Honor Code including prohibitions on plagiarism is in effect for this course. Students are encouraged to talk to each other, to the teaching assistants, to the instructors, or to anyone else about any assignment in the course that is not specifically designated as pledged. *For the whole academic year- signing in another student or having a student sign you in to class will be considered an honor code violation.*

Disability Accommodations

Any student with a disability requiring accommodations in this class is encouraged to contact the instructor after class or during office hours. Additionally, students should contact the Director of Disability Support Services in the Allen Center.

Title IX statement

The University is drafting a statement about Title IX as applied to syllabi. Please know that Rice strives to be an inclusive and supportive environment for all, and that all students and faculty are expected to treat each other with respect at all times.

Privacy

Class and group zoom meetings may be recorded. By default, these meetings won't be made publicly available. We ask senior teams to place some information about their projects online, but this information need not include names or personally identifiable information of any team members. Often, Rice is interested in publishing photographs of interesting projects & students working on them for publicity or pedagogical

purposes. Students will be given an opportunity to sign an optional waiver to cover these situations.

Changes to Syllabus

It should be noted that material in this syllabus is subject to change if announced via Canvas email to the course participants.